

ADVANCE PROBLEM PROGRAMMING

SLO 1: Procedural Concepts

1. Write a Java Function to Calculate Factorial Procedurally

```
public class Main {  
  
    static int factorial(int n) {  
        int fact = 1;  
  
        for (int i = 1; i <= n; i++) {  
            fact = fact * i;  
        }  
  
        return fact;  
    }  
  
    public static void main(String[] args) {  
        int n = 5;  
  
        System.out.println("Factorial = " + factorial(n));  
    }  
}
```

Output:

Factorial = 120

The factorial() function performs the calculation using a loop and returns the result.

2. Differentiate Between Procedure and Method in Java

Procedure

A procedure generally performs an action without returning a value.

It is commonly associated with procedural programming.

It may be described as a void function.

Example: void display()

It may operate mainly on passed parameters or external data.

Method

A method is a block of code defined inside a Java class.

It is a fundamental part of Java's object-oriented programming.

It can return a value or use void.

Example: int add(int a, int b)

It can operate on object data and class data.

Example:

```
class Example {
```

```
    void display() {
```

```
        System.out.println("Hello");
```

```
    }
```

```
    int add(int a, int b) {
```

```
        return a + b;
```

```
    }
```

```
}
```

Here, display() is a void method that performs an action, while add() is a method that returns a value.

3. Identify Drawbacks of Procedural Programming Using a Long Java Function

A long procedural function can become difficult to understand and maintain.

```
public class Main {

    static void processStudent() {

        int marks1 = 80;

        int marks2 = 75;

        int marks3 = 90;


        int total = marks1 + marks2 + marks3;

        double average = total / 3.0;


        if (average >= 50) {

            System.out.println("Pass");

        } else {

            System.out.println("Fail");

        }


        if (average >= 75) {

            System.out.println("Good Performance");

        }


        System.out.println("Total = " + total);

        System.out.println("Average = " + average);

    }


    public static void main(String[] args) {

        processStudent();

    }

}
```

```
}  
  
}
```

Drawbacks of procedural programming:

1. **Difficult to maintain:** Long functions become difficult to modify.
2. **Poor organization:** Data and operations are not grouped into objects.
3. **Code duplication:** The same logic may need to be written repeatedly.
4. **Difficult debugging:** Finding errors in a large function can be time-consuming.
5. **Limited reusability:** Large functions are less flexible for reuse.
6. **Reduced scalability:** Managing large programs becomes difficult as complexity increases.